A PRACTICAL ENGINEERING CASE STUDY

Building and operating a cloud platform

Azure infrastructure. Google Cloud production.
Hands-on deployment, troubleshooting and recovery.

Jeff Cadet

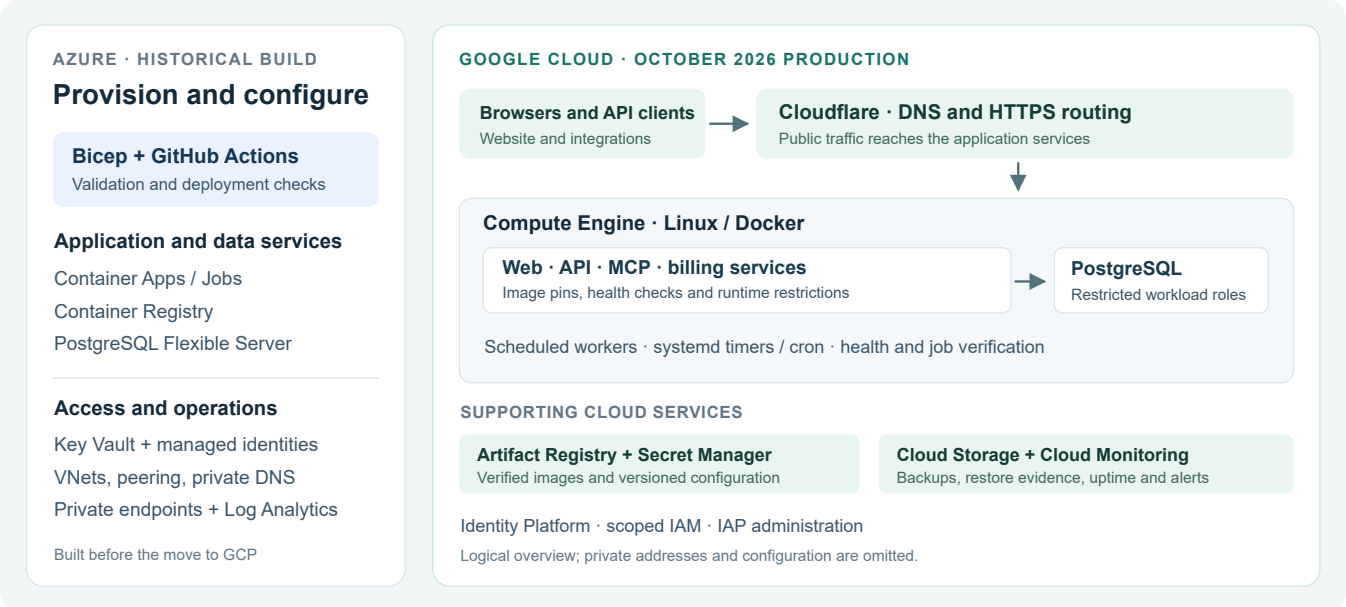
Cloud / Platform Support Engineer · October 2026

I built and supported the cloud environment behind TradeProof, a contractor license and insurance verification platform. My responsibilities covered provisioning, containers, networking, identity, databases, monitoring and production support.

9.5M
License records at the October 8 launch capture

21
CI checks passed for the October 11 release

2 clouds
Azure build and Google Cloud operations



Logical overview of services I worked with; private infrastructure details are omitted. The two clouds represent stages of the project, not an active-active deployment.

> **Project context.** V1.2 was initially built on Azure, then released on Google Cloud on October 8, 2026. The older V1.1 service also ran on GCP; the final production cutover and data reconciliation occurred between those GCP environments.

BUILD, CONFIGURE AND SUPPORT

Work across the platform

From cloud resource configuration to the behavior of the running service.

HISTORICAL ENVIRONMENT

Microsoft Azure

- Built reusable **Bicep** modules and validated deployment changes.
- Deployed **Container Apps and Jobs**, with images in **Azure Container Registry**.
- Provisioned **PostgreSQL Flexible Server** with public network access disabled.
- Configured **VNets, peering, private endpoints and private DNS**.
- Worked with **Key Vault, managed identities and scoped role assignments**.
- Used **Log Analytics** to investigate application and scheduled-job failures.

OCTOBER 2026 PRODUCTION

Google Cloud

- Operated **Linux Compute Engine** hosts and containerized services.
- Published verified Docker images to **Artifact Registry** using immutable references.
- Managed private objects and backups in **Cloud Storage**.
- Maintained versioned recovery configuration in **Secret Manager**.
- Worked with **IAM, Identity Platform and IAP** for service and administrative access.
- Configured **Cloud Monitoring** uptime checks and alerts, with a confirmed delivery test.

DAY-TO-DAY ENGINEERING RESPONSIBILITIES

Container and release support

Built images, checked runtime dependencies, diagnosed startup failures and verified health before activation.

Database and recovery operations

Managed PostgreSQL migrations, permissions, backup verification and restores on isolated infrastructure.

Networking and access

Troubleshot private connectivity, DNS, HTTPS routing, service identity and database permission failures.

Security operations

Applied restricted runtime permissions, staff RBAC, passkey verification and audited access changes.

CI/CD and automation

Maintained GitHub Actions validation, automated API checks, secret scanning and image vulnerability scans.

Documentation and cost control

Maintained runbooks, reviewed compute/storage/CI usage and removed temporary recovery resources after verification.

Tools used: Linux, Docker, PostgreSQL, Bicep, GitHub Actions, Python, Bash, PowerShell, Cloudflare, Trivy and automated browser/API checks.

SYMPTOM → DIAGNOSIS → VERIFIED FIX

Three support investigations

Examples of resolving the failure and checking the resulting behavior.

01 / AZURE Provisioned, but failing at runtime

Symptom: A scheduled Key Vault reminder job failed repeatedly even though the cloud resource reported successful provisioning.

Diagnosis: Log Analytics showed a missing Python module. The Key Vault SDK was installed for development but absent from the runtime image.

Action: Added the runtime dependency, rebuilt the image and verified imports inside the packaged container before deployment.

Verified result: A manual execution succeeded, followed by a successful scheduled run on July 3, 2026. Resource provisioning status alone was insufficient.

02 / GCP A backup upload that hid its failure

Symptom: Cloud Storage contained checksum files without the corresponding database dump files.

Diagnosis: The Cloud SDK could not write local tracking files. The script used the subsequent checksum upload's successful exit status and removed the local dump.

Action: Set a writable SDK configuration directory, checked each upload result and required remote-object verification before deleting local staging files.

Verification: The revised workflow checked actual uploaded artifacts. Later isolated restores separately tested usable backups, database contents and application startup.

03 / RECOVERY Saved configuration no longer matched

Symptom: Recovery configuration had drifted from the deployed services, and a backup manifest still referenced an older, stopped API container.

Diagnosis: Compared running containers and private configuration files with explicit Secret Manager versions and the backup manifest.

Action: Updated recovery versions and the backup reference while retaining the previous versions.

Verified result: All nine configuration files matched their cloud versions. Separate checks confirmed serving services, database permissions and schedules were unchanged.

MEASURED RESULTS, WITH SCOPE

Verify before calling it done

A release process that checks both the artifact and the running workload.

01 · VALIDATE	02 · PREPARE	03 · ACTIVATE	04 · VERIFY
Tests, dependency checks and image scans	Image pins, backups and production canaries	Ordered deployment and health checks	Runtime readback and scheduled-job results

RECORDED RESULT	WHAT WAS VERIFIED
October 8 launch	Full source and target restores, data reconciliation, five healthy application services and public checks before closeout.
21 CI checks passed	October 11 release. The 19 required branch checks and strict policy remained unchanged.
8,879 backend tests	Passed for the same release, with reported test coverage of 89.04%.
Image security checks	Zero fixable HIGH/CRITICAL findings in the exact API and web images under the existing scan policy.
7m 43s recovery drill	September 26 isolated snapshot restore: five apps, public search/navigation and one quota-write check. Authenticated sessions and external-provider recovery were outside this result.

OPERATIONAL DECISIONS THAT MATTERED

The October 8 launch used a planned maintenance window. Once the target accepted production writes, the earlier source remained fenced against writes. A DNS-only reversal could not safely restore the earlier state; recovery required reconciliation.

The October 11 release checked canaries against production configuration, deployed dependencies in order, verified running image identities and observed a successful scheduled worker run. Synthetic API-key checks covered customer authentication behavior without using a live customer's quota.

> **Evidence scope.** Compiled from project task closures, deployment records, CI output and recovery receipts reviewed on October 12, 2026. The figures describe those recorded events. They do not establish audited availability, SOC 2 certification or a guaranteed recovery time. Detailed operational records remain private.

Explore the project

tradeproof.net · Public case study on GitHub

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